SIMULATION EXPERIMENT – 10

❖ **Full-Wave Simulation and Optimization of a Microstrip Patch Antenna for Resonant Frequency, Bandwidth, Gain and Radiation-Pattern Enhancement**

➔ CO Mapping: CO3

Bloom's Level: K6 – Create

WK Level: WK4 – Antenna design and optimization

➔ Objectives of this Experiment:

1. To design a rectangular microstrip patch antenna for a specified microwave operating frequency.
2. To investigate the influence of substrate properties and geometrical dimensions on resonant frequency and impedance matching.
3. To evaluate simulated VSWR, return loss, bandwidth, radiation pattern, gain, directivity and radiation efficiency.
4. To optimize the antenna structure to achieve improved impedance matching and radiation performance.

➔ Expected Outcome from this Experiment:

1. Design and simulate a rectangular microstrip patch antenna for the specified microwave operating frequency.
2. Analyze and interpret the influence of substrate properties and antenna geometrical dimensions on resonant frequency and impedance matching.
3. Evaluate the simulated VSWR, return loss, bandwidth, radiation pattern, gain, directivity and radiation efficiency.
4. Optimize and validate the antenna structure to achieve improved impedance matching and radiation performance based on the specified design requirements and simulated results.

➔ Software Required: CST / Ansys HFSS / ADS

➔ Objective linkage: Design the patch, study substrate/geometrical effects, evaluate VSWR/return loss/bandwidth/radiation performance, and optimize the antenna.

➔ Experimental Procedure:

A. 2.4 GHz Patch Antenna: Design a rectangular microstrip patch antenna on an FR-4 substrate with εr

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= 4.4 and h = 1.6 mm for f0 = 2.4 GHz. Use initial patch dimensions W = 37.2 mm and L = 28.8 mm.

Simulate from 2.0 to 2.8 GHz and obtain resonant frequency, minimum S11, VSWR, −10 dB

bandwidth, gain, efficiency and radiation pattern. Calculate resonant-frequency deviation from 2.4

GHz.

B. Patch Optimization: Change the patch length by ±1 mm and compare the three cases L = 27.8, 28.8

and 29.8 mm. Record resonant frequency, minimum S11, VSWR and bandwidth for each case. Identify

the dimension giving the closest resonance to 2.4 GHz and the best impedance matching, and justify

the selection

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% MICRO10.m - ECA13 Simulation Experiment 10
% 2.4 GHz rectangular microstrip patch antenna (analytical model)
% All results are shown in ONE figure window (2 x 2 subplots).
clear; clc; close all;

c = 3e8;  f0 = 2.4e9;  er = 4.4;  h = 1.6e-3;
tand = 0.02;  Z0 = 50;  Redge = 300;
W = 37.2e-3;

% ---------- Design equations ----------
Wth  = c/(2*f0)*sqrt(2/(er+1));
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eeff = (er+1)/2 + (er-1)/2*(1 + 12*h/W)^(-0.5);
dL   = 0.412*h*((eeff+0.3)*(W/h+0.264))/((eeff-0.258)*(W/h+0.8));
Leff = c/(2*f0*sqrt(eeff));
Ldes = Leff - 2*dL;
fprintf('Theoretical W = %.2f mm\n', Wth*1e3);
fprintf('eeff = %.3f, dL = %.3f mm, Leff = %.2f mm, L = %.2f mm\n', ...
        eeff, dL*1e3, Leff*1e3, Ldes*1e3);

% ---------- Cases: 3 lengths + optimized ----------
f  = linspace(2.0e9, 2.8e9, 1601);
Ls = [27.8e-3, 28.8e-3, 29.8e-3, Ldes];
y0fixed = (28.8e-3/pi)*acos(sqrt(Z0/Redge));
y0opt   = (Ldes/pi)*acos(sqrt(Z0/Redge));
y0s = [y0fixed, y0fixed, y0fixed, y0opt];
dBall = zeros(4, numel(f));
Gall  = zeros(4, numel(f));

fprintf('\n%-8s %-9s %-9s %-7s %-9s %-7s\n', ...
        'L(mm)', 'fr(GHz)', 'S11(dB)', 'VSWR', 'BW(MHz)', 'Dev(%)');
for k = 1:4
    L  = Ls(k);
    fr = c/(2*(L + 2*dL)*sqrt(eeff));
    Qr = c*sqrt(eeff)/(4*fr*h);
    Q  = 1/(1/Qr + tand);
    Rin = Redge*cos(pi*y0s(k)/L)^2;
    Zin = Rin ./ (1 + 1j*Q*(f/fr - fr./f));
    G   = (Zin - Z0) ./ (Zin + Z0);
    dB  = 20*log10(abs(G) + 1e-12);
    Gall(k,:)  = G;
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    dBall(k,:) = dB;

    [gmin, idx0] = min(abs(G));
    fres = f(idx0);
    vswr = (1 + gmin)/(1 - gmin);
    ok   = find(dB <= -10);
    if isempty(ok)
        bw = 0;
    else
        bw = f(ok(end)) - f(ok(1));
    end
    dev = abs(fres - f0)/f0*100;
    fprintf('%-8.2f %-9.4f %-9.2f %-7.2f %-9.1f %-7.2f\n', ...
            L*1e3, fres/1e9, dB(idx0), vswr, bw/1e6, dev);
end

% ---------- Radiation pattern (cavity model) ----------
k0 = 2*pi*f0/c;
th = linspace(-pi/2, pi/2, 721);
Lh = Ldes + 2*dL;
x1 = k0*h/2*sin(th) + eps;
x2 = k0*W/2*sin(th) + eps;
E = abs(cos(k0*Lh/2*sin(th)) .* (sin(x1)./x1) .* cos(th));
H = abs(cos(th) .* (sin(x2)./x2));
EdB = max(20*log10(E/max(E) + 1e-6), -30);
HdB = max(20*log10(H/max(H) + 1e-6), -30);

% ---------- ONE figure with 4 subplots ----------
figure(1);
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set(gcf, 'Position', [100 80 1100 800], 'Color', 'w');

subplot(2,2,1); hold on; grid on;
plot(f/1e9, dBall(1,:), 'b', 'LineWidth', 1.5);
plot(f/1e9, dBall(2,:), 'g', 'LineWidth', 1.5);
plot(f/1e9, dBall(3,:), 'm', 'LineWidth', 1.5);
plot([2.0 2.8], [-10 -10], 'r--');
plot([2.4 2.4], [-40 0], 'k:');
xlabel('Frequency (GHz)'); ylabel('S11 (dB)');
title('(a) S11 for L = 27.8, 28.8, 29.8 mm');
legend('27.8 mm', '28.8 mm', '29.8 mm', 'Location', 'southeast');
ylim([-40 0]);

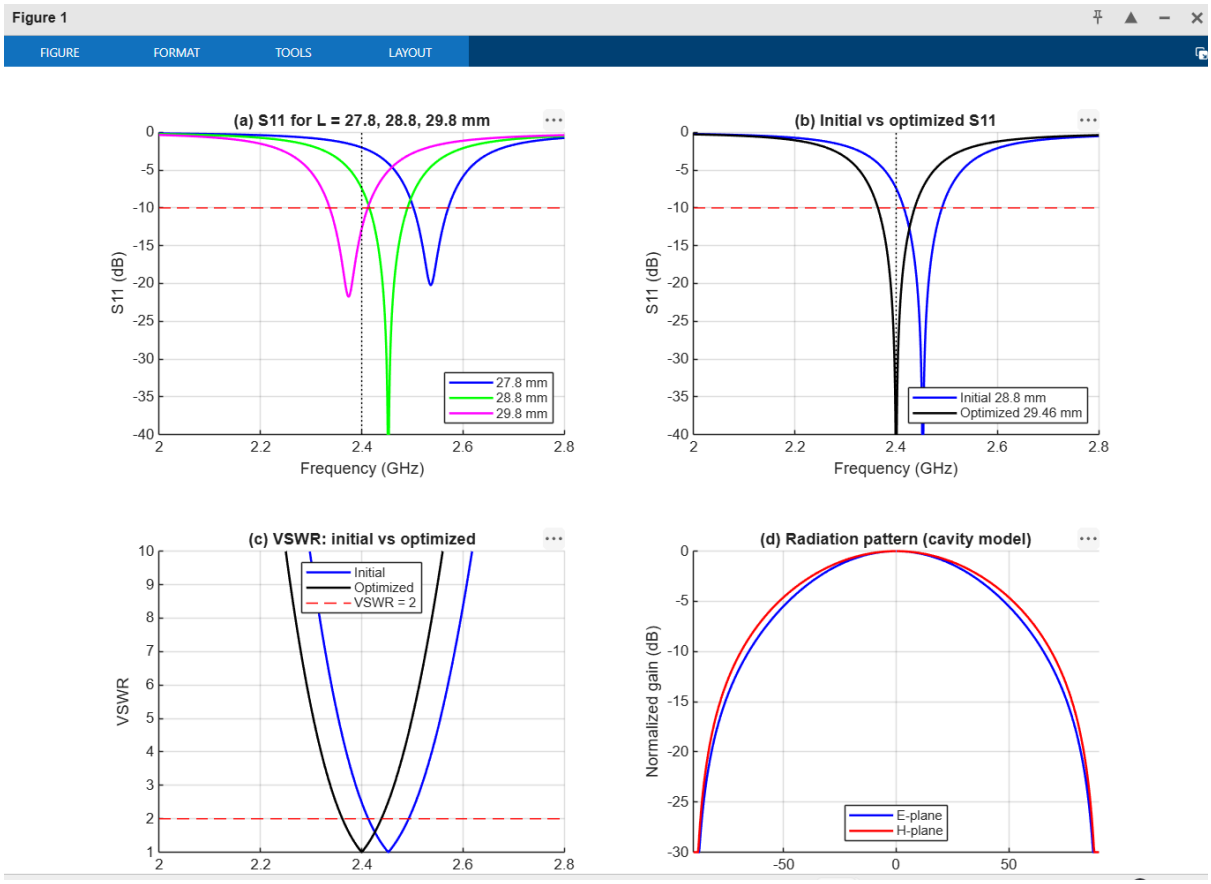
subplot(2,2,2); hold on; grid on;
plot(f/1e9, dBall(2,:), 'b', 'LineWidth', 1.5);
plot(f/1e9, dBall(4,:), 'k', 'LineWidth', 1.5);
plot([2.0 2.8], [-10 -10], 'r--');
plot([2.4 2.4], [-40 0], 'k:');
xlabel('Frequency (GHz)'); ylabel('S11 (dB)');
title('(b) Initial vs optimized S11');
legend('Initial 28.8 mm', sprintf('Optimized %.2f mm', Ldes*1e3), ...
       'Location', 'southeast');
ylim([-40 0]);

subplot(2,2,3); hold on; grid on;
vsI = (1 + abs(Gall(2,:))) ./ (1 - abs(Gall(2,:)));
vsO = (1 + abs(Gall(4,:))) ./ (1 - abs(Gall(4,:)));
plot(f/1e9, vsI, 'b', 'LineWidth', 1.5);
plot(f/1e9, vsO, 'k', 'LineWidth', 1.5);
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plot([2.0 2.8], [2 2], 'r--');
xlabel('Frequency (GHz)'); ylabel('VSWR');
title('(c) VSWR: initial vs optimized');
legend('Initial', 'Optimized', 'VSWR = 2', 'Location', 'north');
ylim([1 10]);

subplot(2,2,4); hold on; grid on;
plot(th*180/pi, EdB, 'b', 'LineWidth', 1.5);
plot(th*180/pi, HdB, 'r', 'LineWidth', 1.5);
xlabel('Theta (deg)'); ylabel('Normalized gain (dB)');
title('(d) Radiation pattern (cavity model)');
legend('E-plane', 'H-plane', 'Location', 'south');
xlim([-90 90]); ylim([-30 0]);

print(gcf, 'MICRO10_all_figures.png', '-dpng', '-r200');
```



Theoretical W = 38.04 mm
eeff = 4.081, dL = 0.739 mm, Leff = 30.94 mm, L = 29.46 mm

L(mm)	fr(GHz)	S11(dB)	VSWR	BW(MHz)	Dev(%)
27.80	2.5365	-20.20	1.22	68.0	5.69
28.80	2.4525	-77.54	1.00	74.5	2.19
29.80	2.3740	-21.73	1.18	74.5	1.08
29.46	2.4000	-240.00	1.00	72.5	0.00

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